MIL-E-20652F
28 March 1986
SUPERSEDING
MIL-E-20652E
31 March 1972

MILITARY SPECIFICATION

EYELETS, METALLIC, AND EYELET WASHERS, METALLIC, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the general requirements for commercial metallic eyelets and washers.

1.2 Classification. The types, sizes, materials, and finishes shall be as designated on the detail document sheet which is part of this general document.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5325

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SPECIFICATIONS

FEDERAL

P-S-1792	-	Soap, Laundry (Neutral and Built)
QQ-B-613	-	Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-F-495	-	Finish, Chemical, Black, for Copper Alloys
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on the date of invitation for bids or request for approval, from a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 Specification sheets. The individual detailed specifications shall be as specified herein and in accordance with the applicable specification sheets.

3.2 Materials and components. The materials and components shall be as specified herein. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed end item complies with all the provisions of this document.

3.2.1 Brass. Brass shall conform to copper alloy numbers 260 or 268 sheet or strip of QQ-B-613. Brass for eyelets shall be annealed suitable for forming. Brass for flat washers shall be tempered, minimum half-hard.

3.2.2 Aluminum. Aluminum for eyelets and washers shall be of the quality used by the manufacturer for eyelets.

3.3 Design and construction. The design and construction of the eyelets and washers shall be as specified herein and as shown on the applicable specification sheets. Eyelets metal shall not crack when clinched as specified in 4.2.2.3.

3.4 Finish.

3.4.1 Enamel finish.

3.4.1.1 Pretreatment. Aluminum eyelets and washers that require an enamel finish shall be given a chemical treatment in accordance with class 1A of MIL-C-5541, prior to the application of the enamel. Brass eyelets that require enameling shall have a dark oxide pretreatment applied in accordance with commercial practice and shall be thoroughly rinsed before enameling.

3.4.1.2 Enameling. Following pretreatment, eyelets and washers requiring enamel shall be given two coats of the color enamel as specified in the contract or purchase order (see 6.2). Enamel shall be applied and baked. The dried enamel shall be semi-gloss. The white color shall approximate No. 27875 of FED-STD-595. The enamel shall show not more than a slight discoloration and shall resist thumbnail penetration when tested for resistance to hot soap solution as specified in 4.3.1. The enamel shall not crack or flake when tested as specified in 4.2.2.3 for clinching.

3.4.2 Black chemical finish. Brass eyelets and washers requiring a black chemical finish shall be finished in accordance with MIL-F-495.

3.4.3 Anodized finish. Anodized aluminum eyelets shall have a non-dyed (natural) anodic coating.

3.4.4 Brass natural finish. Natural brass finish shall be as normally furnished by the supplier.

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3.5 Workmanship. The finished eyelets and washers shall be free from fracture, malformation, burrs, and rough or sharp edges. Flat washers shall be smooth and free of any bend. Finish shall be smooth and free of foreign imbedded matter, any discoloration, and areas of no film.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicted or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

4.2.1.1 In-process inspection. Examination shall be made to determine compliance with the finish preparation requirements in 3.3.1.1, and the application of enamel requirements in 3.3.1.2. Whenever noncompliance is noted, correction shall be made to the applicable process and all items processed.

4.2.2 End item inspection.

4.2.2.1 End item visual examination. The end item shall be examined for defects listed in table I. The lot size shall be expressed in units of one eyelet or one washer. The sample unit shall be one eyelet or one washer. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

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TABLE I. End item visual defects

Examine	Defect
Finish (eyelet and washer)	Type of finish or color not as specified Area of no film (when applicable), not smooth, foreign matter imbedded, discoloration
Design, construction and workmanship (eyelet and washer)	Not specified type or size Fracture or malformation Burr, rough or sharp edge Bent

4.2.2.2 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified. Any dimension not within the specified tolerance as specified on the applicable specification sheet shall be classified as a defect. The lot size shall be expressed in units of one eyelet or one washer. The sample unit shall be one eyelet or one washer. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.2.3 End item testing. The end item shall be tested for the characteristics listed in table II. The lot size shall be expressed in units of one eyelet or one washer. The sample unit shall be three eyelets or three washers, as applicable. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE II. End item testing

Characteristic	Requirement	Test method	Number of determinations per sample unit	Result reported
Metal resistance to clinching	3.3	4.3.2	1	Pass or fail
Enamel finish:				
Resistance to hot soap solution and fingernail	3.4.1.2	4.3.1	1	Pass or fail
Resistance to clinching	3.4.1.2	4.3.2	1	Pass or fail

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4.2.3 Packaging examination. The fully packaged end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL expressed in terms of defects per hundred units, shall be 2.5.

TABLE III. Packaging defects

Examine	Defect
Marking (exterior and interior)	Missing; incorrect; incomplete; illegible, of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Bulged or distorted container
Contents (exterior and interior container)	Weight exceeds specified amount Quantity is more or less than indicated (see 5.4.1)

4.2.4 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed in table IV. The lot size sahll be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE IV. Palletization defects

Examination	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Palletized pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limit
Marking	Omitted; incorrect; illegible; of improper size location, sequence, or method of application

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4.3 Methods of inspection.

4.3.1 Enamel resistance to hot soap solution and thumbnail test. The enameled eyelet or washer under test shall be immersed in a 5 percent soap solution maintained at 195°F, plus or minus 5°F, for 2 hours. The soap shall conform to P-S-1792. The eyelets or washers shall be rinsed and the enamel on the top examined for differences in appearance from an unimmersed control specimen and an attempt made to furrow through the film with the thumbnail. Any non-conformance with the applicable enamel requirements specified in 3.3.1.2 shall constitute failure of this test.

4.3.2 Clinching test. An eyelet shall be placed in an eyeleting machine and clinched onto material usually used with eyelets. Any evidence of metal cracks or flaking of enamel shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), eyelets or washers of one size only shall be unit packed separately, not to exceed 20 pounds, in a snug-fitting fiberboard box conforming to style RSC, type CF (variety optional) or SF, class domestic, grade 200 of PPP-B-636. Each box shall be closed in accordance with the appendix of the box specification.

5.1.2 Commercial. Eyelets and washers shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level packing. Unless otherwise specified (see 6.2), eyelets or washers of one size only preserved as specified in 5.1, shall be packed separately, not to exceed 120 pounds, in a snug-fitting shipping container conforming to style FTC, type SF, class weather-resistant, grade V2s of PPP-B-636, except that the weight limitations shall be waived. Each shipping container shall be closed and reinforced with strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B packing. Unless otherwise specified (see 6.2), eyelets or washers of one size only preserved as specified in 5.1, shall be packed separately, not to exceed 120 pounds, in a snug-fitting double-walled, fiberboard shipping container conforming to style FTC, type CF, variety DW, class domestic, grade 350 of PPP-B-636. Each shipping container shall be closed and reinforced as specified in 5.2.1.

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5.2.2.1 Weather-resistant container. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fabricated and closed in accordance with PPP-B-636.

5.2.3 Commercial packing. Eyelets and washers, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), eyelets and washers, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type III, Type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the container patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Additional marking. Each interior unit pack and shipping container shall be marked with the approximate quantity of items contained therein.

6. NOTES

6.1 Intended use. The eyelets and washers are intended to be used on various military items such as clothing, footwear, tentage, and equipage and for other applications.

6.2 Ordering data. Acquistion documents should specify the following:

a. Title, number, and date of this document.
b. Basic number and dash number of eyelet and washer required. (For telescopic eyelets, specify two numbers i.e., eyelet plus washer, see MIL-E-20652/3).
c. Material required (see 3.2.1, 3.2.2, and MIL-E-20652/1,/2 and/3.)
d. Finish required (see 3.3.1, 3.3.2, 3.3.3, 3.3.4, and MIL-E-20652/1 and/3.)
e. Color of enamel required (see 3.3.1.2, MIL-E-20652/1 and/3).
f. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
g. When quantity per unit pack and pack is other than specified (see 5.1.1, 5.2.1 and 5.2.2).
h. When weather-resistant fiberboard shipping containers are required for level B packing (see 5.2.2.1).
i. When palletization is required (see 5.3).

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6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - AR
Air Force - 82
DLA - IS

User activities:

Army - AT, ME
Navy - MC

Preparing activity:

Army - GL

Project No. 5325-0224

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-20652F	Eyelets, Metallic, and Eyelet Washers, Metallic, General Specification For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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